



NATIONAL INSTITUTE OF FASHION TECHNOLOGY ,BHUBANESWAR

Department Of Fashion Technology

Apparel Merchandising & Sourcing and Costing

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OBJECTIVES

- To study the apparel manufacturing industry of Turkey and Morocco
- To analyze sourcing ecosystems and export competitiveness
- To compare heavyweight hoodie manufacturing capabilities
- To study fabric, trims, printing, and production systems
- To develop comparative FOB costing analysis
- To analyze future opportunities and industrial challenges



GLOBAL APPAREL INDUSTRY OVERVIEW

- The global apparel industry is one of the largest manufacturing sectors worldwide.
- The industry is driven by fast fashion, athleisure, and streetwear demand. Europe and the United States remain major apparel importing markets.
- Oversized streetwear garments have gained popularity due to Gen Z fashion trends.
- Nearshoring and sustainable sourcing are reshaping global supply chains.

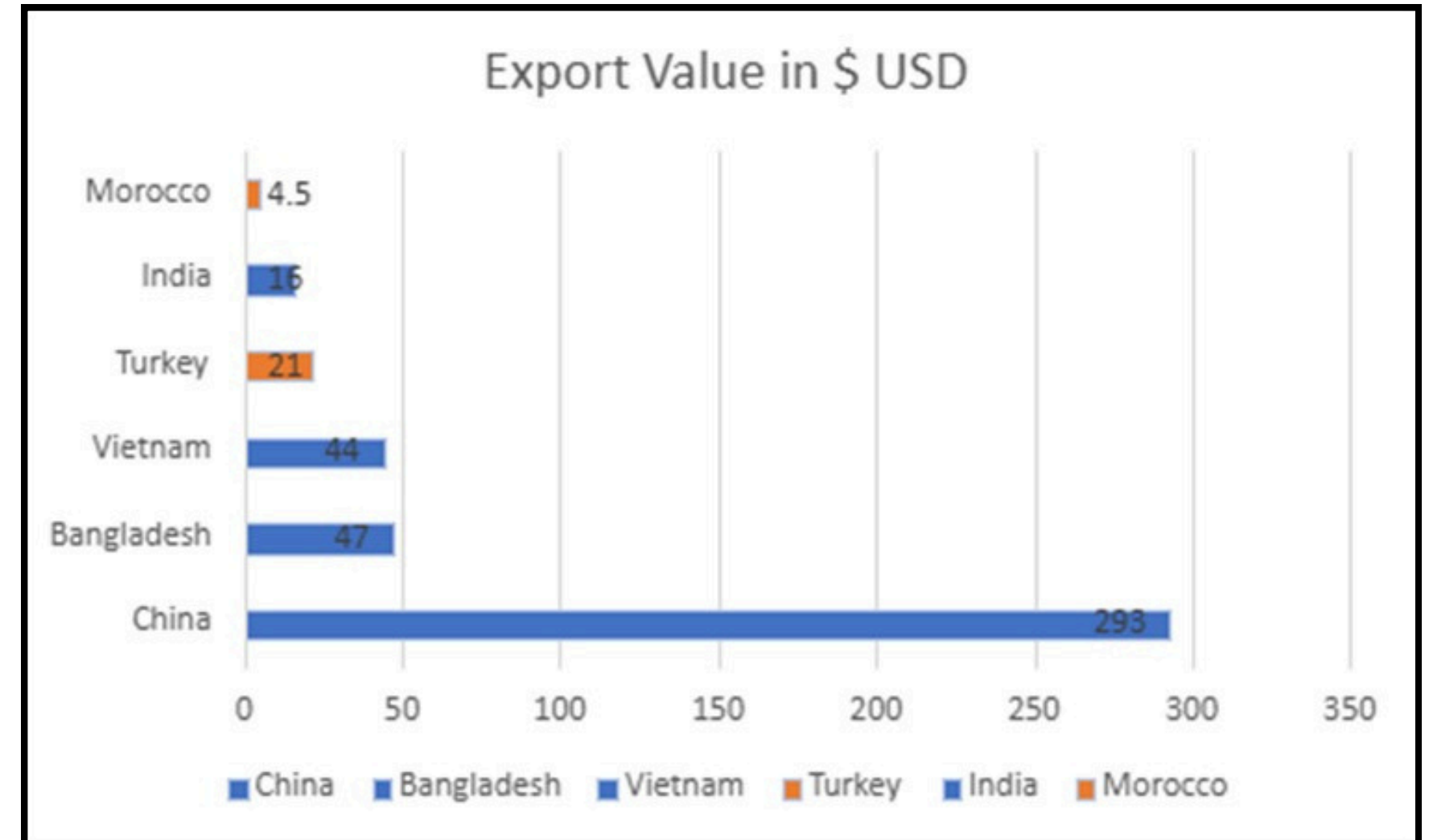


Figure 1 Approximate apparel export values based on WTO, UN Comtrade, national trade bodies, and industry reports (2024–2025).

GLOBAL IMPORT REGIONS & SOURCING

- The European Union is the world's largest apparel importing region. Fast fashion and premium streetwear brands rely heavily on nearshore sourcing. Turkey and Morocco play a major role in supplying the EU market. Shorter lead times and lower logistics costs are increasing nearshore demand.

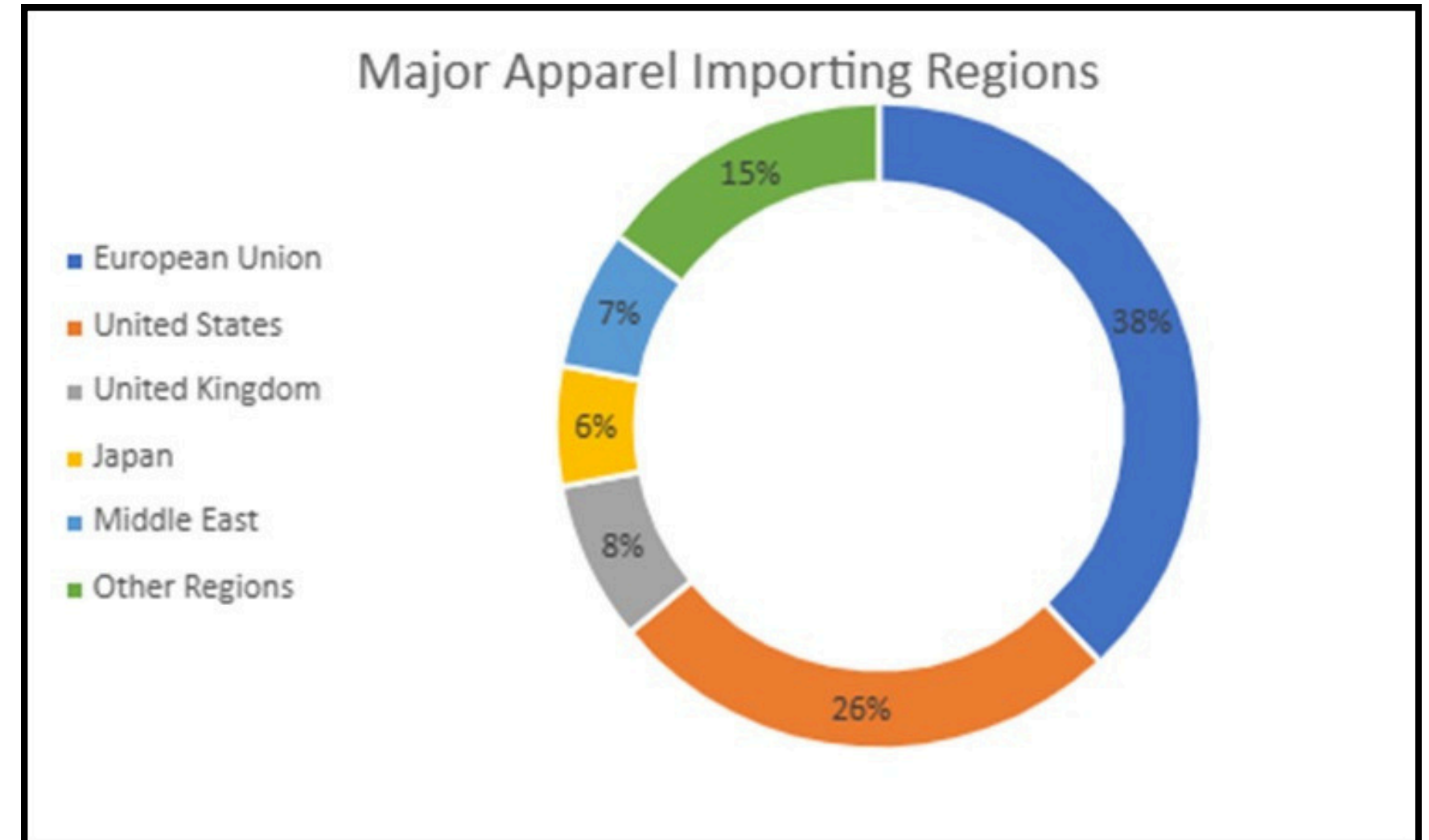


Figure 2 Approximate global apparel import share distribution based on OECD sourcing trends, WTO trade reports, McKinsey fashion market analysis, and international apparel consumption patterns (2024–2025).

WHY TURKEY & MOROCCO WERE SELECTED

WHY TURKEY?

Turkey is one of the world's largest knitwear exporters. Knitwear contributes approximately 52% of Turkey's apparel exports. Turkey exported approximately USD 21 billion worth of apparel products in 2025. Turkey has a vertically integrated textile ecosystem:

- Turkey is highly preferred for: premium streetwear heavyweight fleece sustainable apparel manufacturing luxury casualwear Major buyers: Zara, H&M, Mango, Bershka, Pull&Bear, Nike, Adidas



WHY TURKEY & MOROCCO WERE SELECTED

WHY MOROCCO?

Morocco is one of Europe's largest nearshoring apparel hubs. Morocco exported approximately USD 4.5 billion apparel products. Over 1,600 apparel manufacturing companies operate in Morocco.

Morocco specializes in:

- fast-fashion replenishment
- low MOQ production
- short lead-time manufacturing
- EU-focused sourcing

Morocco's strategic advantage: Tanger Med Port enables 2–3 day EU delivery.



Tanger Med Port enables 2–3 day EU delivery.

TURKEY AND MOROCCO APPAREL INDUSTRY INDICATORS

Parameter	Turkey	Morocco
Apparel Export Value	USD 21B	USD 4.5B
Apparel Workforce	1 Million	220,000
Knitwear Share	52%	35%
Lead Time to EU	42 - 45 Days	45-52 Days

MAJOR PRODUCTS MANUFACTURED IN TURKEY & MOROCCO

TURKEY — MAJOR APPAREL PRODUCTS

Knitwear

- Sweatshirts
- Hoodies
- T-shirts
- Sportswear

Joggers

Denim

Premium casualwear

- Knitwear accounts for approximately 52% of Turkish apparel exports.
- Sweatshirts and hoodies are among Turkey's fastest-growing knitwear categories.

MOROCCO — MAJOR APPAREL PRODUCTS

Knitwear

- Hoodies
- Basic knitwear
- Sportswear
- Sweatshirts

Denim

Trousers

Fast-fashion tops

- Fast-fashion apparel contributes approximately 52% of Morocco's textile production.
- Morocco is highly specialized in EU replenishment manufacturing.

PRODUCT CATEGORY DISTRIBUTION IN TURKEY AND MOROCCO

APPAREL INDUSTRY

Product Category	Turkey	Morocco
Knitwear	52%	35%
Denim	18%	22%
Sportswear	15%	18%
Fast Fashion Basics	15%	25%

PRODUCT CATEGORY: MEN'S HOODIE

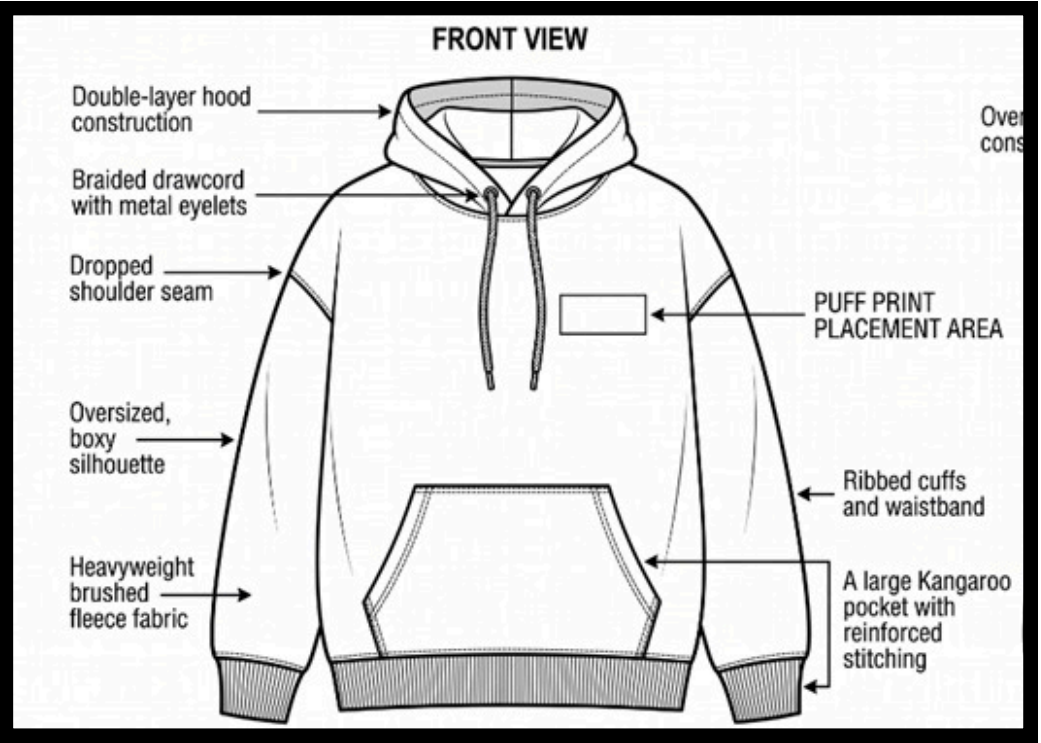
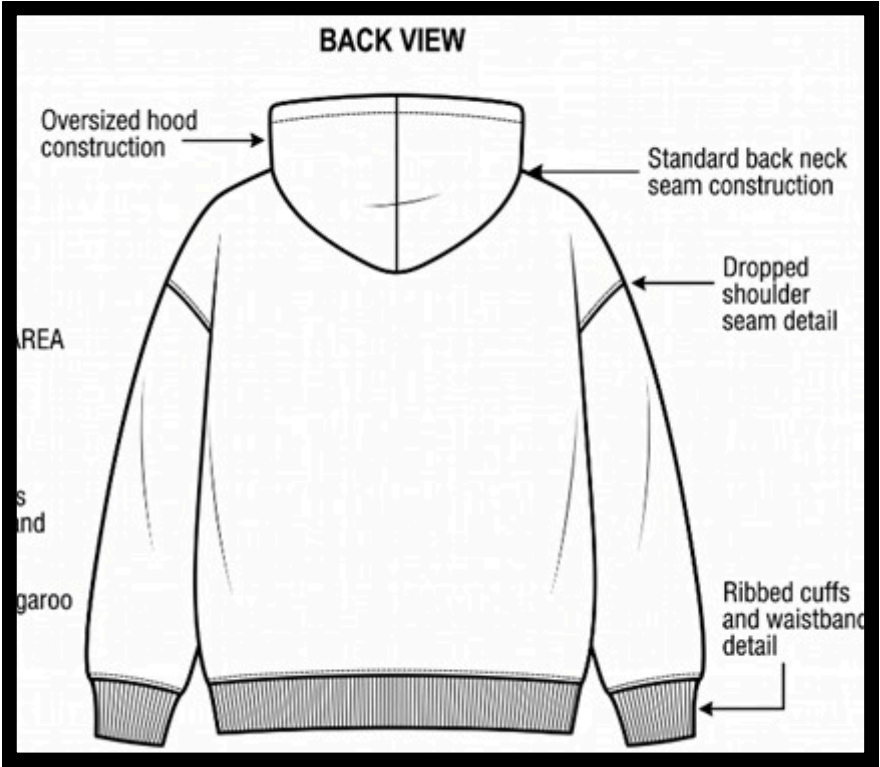
GLOBAL STREETWEAR MARKET

- Streetwear market valued at approximately USD 185 Billion.
- Oversized silhouettes dominate Gen Z fashion trends.
- Heavyweight hoodies are key products in:
 1. Streetwear
 2. Athleisure
 3. Casual wear

WHY HOODIES ARE COMMERCIALLY IMPORTANT

- High year-round demand
- Unisex market adaptability
- High retail margins
- Premium fabric compatibility

TECHNICAL FLAT SKETCH + SPECIFICATIONS



AVAILABLE HOODIE STYLES IN MARKET

COMMON COMMERCIAL HOODIE TYPES

Hoodie Type
Pullover Hoodie
Zip-Up Hoodie
Cropped Hoodie
Oversized Hoodie
Heavyweight Hoodie
Graphic Hoodie



WHY OVERSIZED HEAVYWEIGHT HOODIE WAS SELECTED

- Highly commercial
- Strong European demand
- Widely manufactured in Turkey
- Increasingly produced in Morocco

SUITABLE FOR:

- puff printing
- enzyme washing
- heavyweight fleece

SELECTED PRODUCT STYLE

Men’s Oversized Heavyweight Hoodie

PRODUCT SPECIFICATIONS

Parameter	Specification
Product	Men’s Hoodie
Fit	Oversized
GSM	350–360 GSM
Fabric	80/20 Cotton-Poly Fleece
Hood	Double Layer
Pocket	Kangaroo Pocket
Rib	2×2 Rib
Print	Puff Print
Wash	Silicone + Enzyme

COMMERCIAL CHARACTERISTICS

- Boxy silhouette
- Dropped shoulders
- Heavyweight drape
- Premium hand-feel
- Streetwear-oriented construction

FABRIC ANALYSIS

SELECTED FABRIC:
80/20 Cotton-Polyester Brushed Fleece
(350–360 GSM)

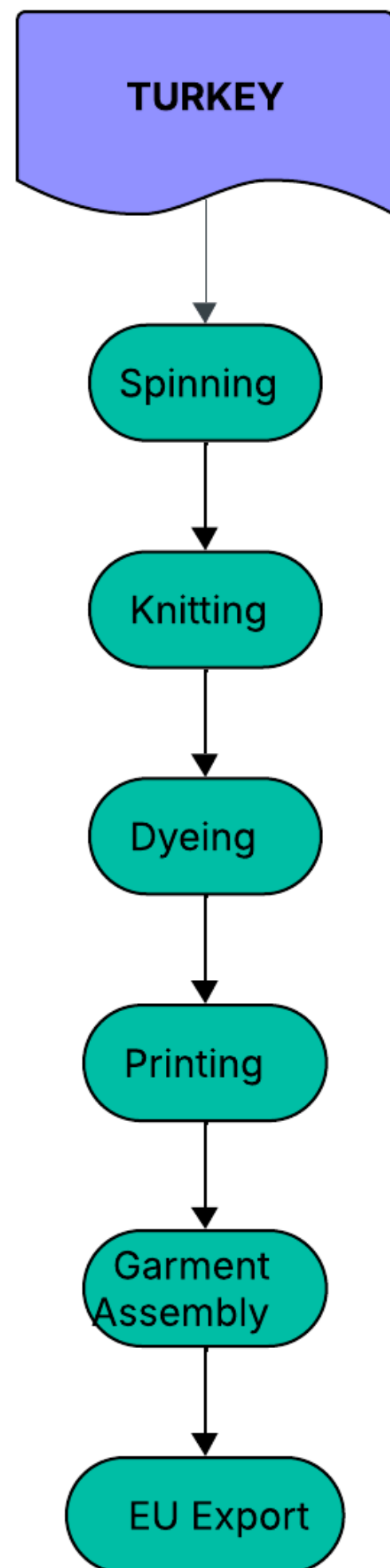
FABRIC CONSTRUCTION

Component	Yarn Type
Face Yarn	30s Combed Cotton
Tie Yarn	Polyester Filament
Back Yarn	Carded Cotton Blend

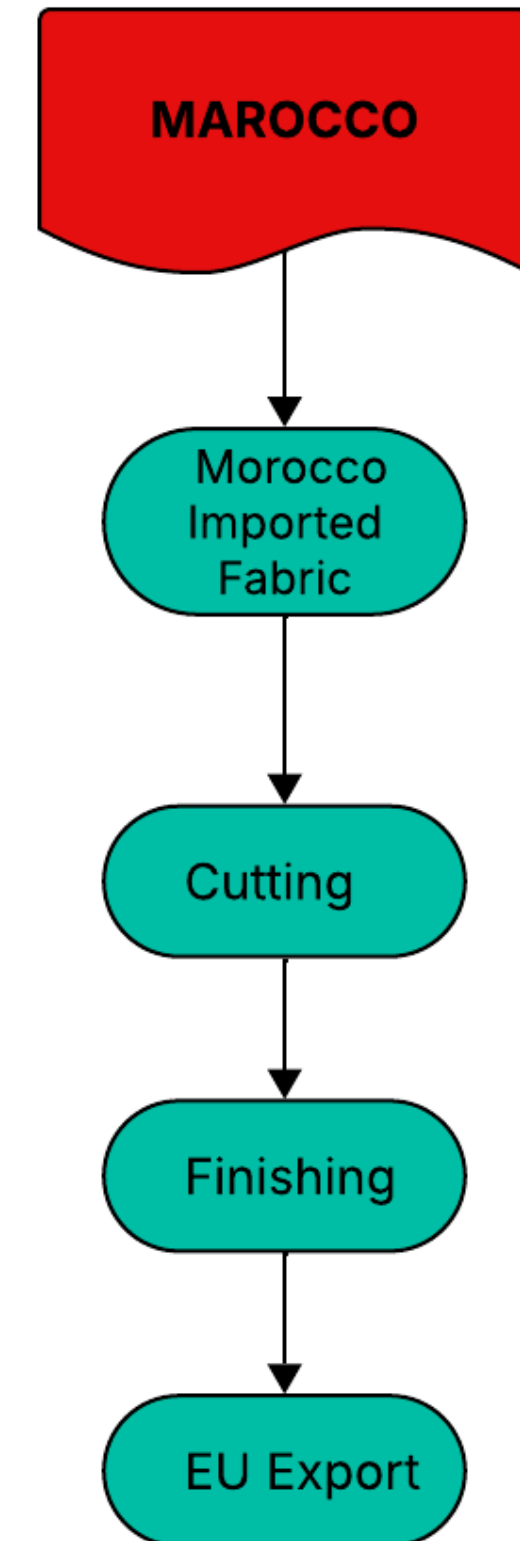


WHY THIS FABRIC WAS SELECTED

- High durability
- Better shape retention
- Reduced shrinkage
- Improved printability
- Better thermal insulation
- Commercial availability
- Ideal for oversized silhouettes



FABRIC MANUFACTURING PROCESS



FABRIC PERFORMANCE & TESTING

Property	Standard
GSM	350–360
Shrinkage	≤3%
Pilling	Grade 4–5
Color Fastness	Grade 4–5
Bursting Strength	≥450 kPa

FABRIC SOURCING (TURKEY)

MAJOR MANUFACTURING REGIONS

- Kahramanmaraş
- Bursa
- Istanbul
- Denizli

MAJOR SUPPLIERS

- Yeşim Tekstil
- İhya Tekstil
- Cemsa Textile
- Egedeniz Textile

STRENGTHS

- Vertical integration
- Local cotton sourcing
- Advanced finishing
- Sustainable certifications



FABRIC SOURCING (MOROCCO)

IMPORT DEPENDENCY

Heavy reliance on imported fleece fabric

Main Import Sources

- Turkey
- Portugal
- Spain

Major Suppliers

- Kiron SA
- Dounitex
- Jinane Fashion



TRIMS ANALYSIS

MAIN TRIMS

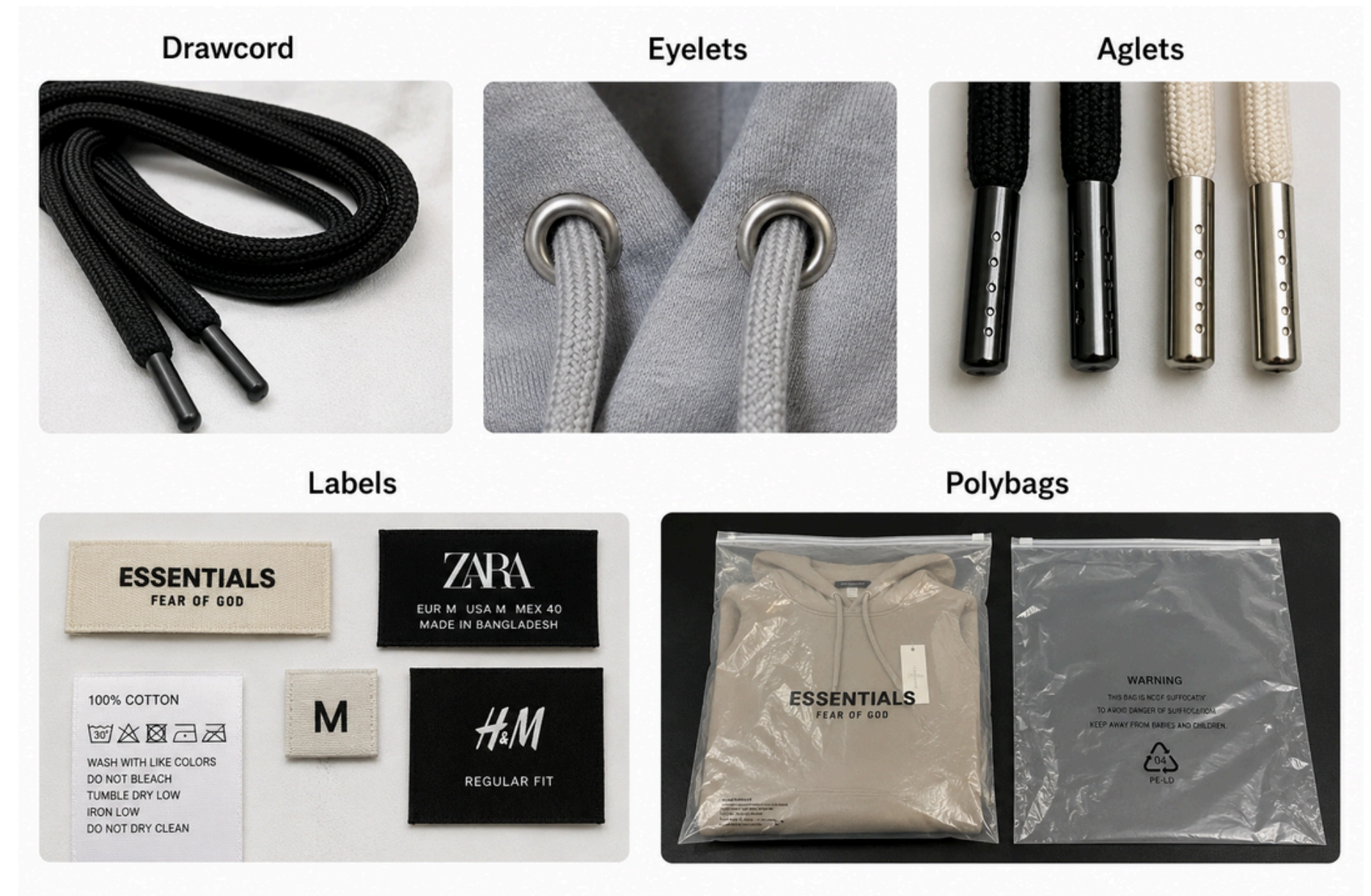
- Drawcord
- Eyelets
- Aglets
- Labels
- Polybags

TECHNICAL SPECIFICATIONS

- Brass eyelets
- Cotton braided cords
- Recycled LDPE packaging

MAJOR SUPPLIERS

- Ata Buttons
- Trimist



PRINTING & FINISHING

PUFF PRINT

- 3D texture
- streetwear effect
- Heat-expanded plastisol print

SILICONE ENZYME WASH

- Improved softness
- Reduced pilling
- Enhanced drape

PUFF PRINT

- 3D texture
- streetwear effect
- Heat-expanded plastisol print



SILICONE ENZYME WASH

- Improved softness
- Reduced pilling
- Enhanced drape



INDUSTRIAL LABOUR COSTING ANALYSIS

Product	Men's Oversized Heavyweight Hoodie	
Country	Turkey	Morocco
Buyer Category	European Fast Fashion	European Fast Fashion
Fabric	80/20 Cotton Polyester Brushed Fleece	80/20 Cotton Polyester Brushed Fleece
MOQ	1,500 Garments	3,000 Garments

Size	Ratio
S	1
M	2
L	2
XL	1

S.No	Operation	Machine Type	Stitch Class	SAM
1	Shoulder Join	Overlock	504	0.45
2	Hood	Overlock	504/514	0.6
3	Hood Attach	Flatlock	607	0.6
4	Sleeve Attach	Flatlock	607	0.85
5	Side Seam Join	Overlock	504	0.7
6	Pocket Hem Fold	SNLS	301	0.4
7	Pocket Attach	SNLS	301	0.6
8	Rib Join	Overlock	504	0.35
9	Waistband Attach	Coverstitch	602	0.55
10	Bottom Hem Finish	Coverstitch	605	0.35
11	Eyelet Attach	Eyelet Press	—	0.4
12	Label Attach	SNLS	301	0.25
13	QC & Trimming	Manual	—	0.5
14	Pressing	Steam Vacuum	—	1.2
15	Packing	Packing Table	—	1.5

Allowance	SAM
Bundle Handling	1.2
Machine Delay	1.1
Operator Fatigue	0.9
Inline QC Delay	0.75
Panel Matching	0.95
Puff Print Handling	0.8
Washing Rechecking	1
Total sam	16

Factory Labour Assumptions	Turkey	Moroco
Particulars	Value	Value
Total Sewing Operators	35	33
Working Days/Month	26	26
Working Hours/Day	8	8
Minutes/Day	480	480
Factory Efficiency	80%	78%

Total Available Minutes	Turkey	Morocco
	436800	411840
Total available min *factory efficiency	349440	321235.2

Total Labour Cost per month	Turkey	Morocco
Salary for a single labour	\$600	\$300
Total labour salary	\$21,000	\$9,900
Then additional: helpers, supervisors, cutting support, QC staff, finishing support	\$27,000	\$13,500

CPM Calculation	Turkey	Morocco
Total Labour Cost per month/Total Available minutes	0.077266484	0.042025282
approx.	0.08	0.05
Base cm(sewing opertor)	1.28	0.8
addition of cutting support,trimming,checking,packing.	\$2.08	\$1.46

DIRECT MATERIAL COST TURKEY AND MOROCCO

TURKEY				MOROCCO		
MOQ:1500				MOQ:3000		
Cutting	Cost/Garment	Total Cost		Particulars	Cost/Garment	Total Cost
Fleece Fabric	\$8.63	\$12,945		imported	\$9.10	\$27,300
Rib Fabric	\$1.79	\$2,685		Rib Fabric	\$1.95	\$5,850
Drawcord	\$0.35	\$525		Drawcord	\$0.48	\$1,440
Eyelets	\$0.04	\$60		Eyelets	\$0.08	\$240
Aglets	\$0.10	\$150		Aglets	\$0.12	\$360
Sewing Thread	\$0.18	\$270		Sewing Thread	\$0.18	\$540
Labels	\$0.13	\$195		Labels	\$0.16	\$480
Packaging	\$0.23	\$345		Packaging	\$0.28	\$840
Total	\$11.45	\$17,175		Total	\$12.35	\$37,050

DIRECT LABOUR COST

DIRECT LABOUR COST				
Turkey			Morocco	
Department	Total Cost		Department	Total Cost
Cutting	\$210		Cutting	\$300
Sewing	\$1,890		Sewing	\$2,760
Finishing & QC	\$540		Finishing & QC	\$720
Packing Department	\$480		Packing Department	\$600
Total Labour Cost	\$3,120		Total Labour Cost	\$4,380
Labour Cost/Garment	\$2.08		Labour Cost/Garment	\$1.46

DIRECT EXPENSES & PRIME COST

Particulars	Cost/Garment	Total Cost		Particulars	Cost/Garment	Total Cost
Puff Print	\$0.85	\$1,275		Puff Print	\$0.90	\$2,700
Silicone + Enzyme Wash	\$0.70	\$1,050		Silicone + Enzyme Wash	\$0.80	\$2,400
Fabric Testing	\$0.10	\$150		Fabric Testing	\$0.12	\$360
Total Direct Expense		\$2,475		Total Direct Expense		\$5,460
Turkey				Morocco		
The total of all direct costs involved	\$22,770.00			The total of all direct costs involved	\$46,890.00	

OVERHEADS

	Turkey	Morocco
Overhead	\$20,295	\$41,430
Overhead 15%	\$3,044	\$6,215

INDIRECT COST

Indirect cost		
Particulars	Total Cost(TURKEY)	Total Cost(MOROCCO)
Bundle Tags	\$45	\$90
Barcode Stickers	\$30	\$60
Carton Markings	\$15	\$30
Carton Tape Allocation	\$23	\$45
Polybag Stickers	\$15	\$30
Assortment Sheets	\$30	\$60
Total Indirect Material Cost	\$158	\$315

LOGISTICS & EXPORT COST

LOGISTICS & EXPORT COST		
Particulars	Total Cost(TURKEY)	Total Cost(MOROCCO)
Inland Transportation	\$450	\$720
Export Documentation	\$300	\$420
Customs Coordination	\$180	\$240
Port Handling & Loading	\$420	\$690
Container Handling	\$210	\$390
Cargo Movement to	\$150	\$240
Total Logistics Cost	\$1,710	\$2,700

	TURKEY	MOROCCO
FOB cost	\$27,682.25	\$56,119.50
profit of 8%	\$2,768.23	\$4,489.56
final fob value	\$30,450.48	\$60,609.06
cost per garment	\$20.30	\$20.20

**TURKEY: TIME & ACTION
CALENDAR**



COMPARATIVE EXPORT, MANUFACTURING, COSTING & LOGISTICS ANALYSIS

Turkey vs Morocco Apparel Industry Comparison

EXPORT, MANUFACTURING

Parameter	Turkey	Morocco
Apparel Export Value	~\$21 Billion	~\$4.5 Billion
EU Export Dependency	High	Very High
Knitwear Strength	Very Strong	Moderate
Major Export Markets	Germany, Spain, UK	Spain, France
Manufacturing Model	Vertical Integration	Nearshore CMT

COSTING

Cost Parameter	Turkey	Morocco
Fabric Cost	Lower	Higher
Labour Cost	Higher	Lower
Trim Cost	Lower	Slightly Higher
Printing Cost	Moderate	Moderate–High
Logistics Cost	Moderate	Low
FOB Cost	~\$20.30	~\$20.20
Production Model	Fast Fashion	Fast Fashion

LOGISTICS

Parameter	Turkey	Morocco
Total Lead Time	42–45 Days	45–52 Days
Fabric Lead Time	Low	High
Shipment Transit to EU	3–4 Days	2–3 Days
Main Export Method	Road / Seaways	Seaways
Major Port	Ambarlı	Tanger Med
Replenishment Speed	High	Very High

SWOT ANALYSIS: TURKEY

Strengths	Weaknesses
Strong vertical integration	Higher labor costs
Premium knitwear expertise	Inflation volatility
Sustainable textile ecosystem	Higher operational costs
Advanced finishing technology	Energy price fluctuations

Opportunities	Threats
Growth in premium streetwear	Competition from Asia
Organic cotton expansion	Currency instability
Sustainable apparel demand	Rising production costs
EU proximity advantage	Global sourcing diversification

PESTEL ANALYSIS: TURKEY

Factor	Analysis
Political	Strong EU trade relations
Economic	Inflation and currency volatility
Social	Skilled textile workforce
Technological	Advanced knitting and dyeing
Environmental	Sustainability regulations increasing
Legal	Compliance with EU textile

KEY INSIGHT
Turkey competes through premium manufacturing and vertical integration.

SWOT ANALYSIS: MOROCCO

Strengths	Weaknesses
Low labor cost	Import dependency
Fast EU shipping	Limited textile integration
Nearshore manufacturing	Limited premium finishing
Strong fast-fashion capability	Fabric sourcing dependency

Opportunities	Threats
Growth in EU fast fashion	Asian sourcing competition
Expansion of D2C production	Logistics cost fluctuations
Sustainable transformation	Textile trade imbalance
Increased EU sourcing shift	Energy dependency

PESTEL ANALYSIS: MOROCCO

Factor	Analysis
Political	Strong EU trade agreements
Economic	Competitive labor cost
Social	Growing apparel workforce
Technological	Expanding automation adoption
Environmental	Green transformation initiatives
Legal	EU-compliant export manufacturing

KEY INSIGHT
Morocco’s competitive advantage lies in nearshore fast-fashion sourcing.

LABOUR FORCE ANALYSIS

Turkey

- Apparel workforce exceeds approximately 1 million workers.
- Strong technical textile skill base.
- Advanced knitwear manufacturing expertise.
- Higher labour productivity.

Morocco

- Apparel workforce approximately 220,000 workers.
- Competitive labor-intensive production.
- Strong fast-fashion assembly capability.
- Lower CMT labor cost.

Parameter	Turkey	Morocco
Workforce	~1 Million	~220,000
Monthly Wage	Higher	Lower
Manufacturing Focus	Premium	Fast Fashion
Productivity	High	Moderate

CITATION
(İHKİB, 2025; AMITH, 2025)

CHALLENGES OF THE GARMENT INDUSTRY

Turkey Challenges

- Rising labor cost
- Inflation pressure
- Energy cost increase
- Currency fluctuations

Morocco Challenges

- Heavy fabric import dependency
- Limited textile integration
- Supply chain vulnerability
- Dependence on the EU market

Global Challenges

- Fast-changing fashion trends
- Sustainability compliance costs
- Competitive pressure from Asia
- Shorter lead-time expectations

CONCLUSION

- This study compared the sourcing, manufacturing, and FOB costing of Men's Oversized Heavyweight Pullover Hoodies in Turkey and Morocco.
- Turkey is a vertically integrated apparel manufacturing hub known for premium streetwear, sustainable production, heavyweight fleece manufacturing, and advanced textile capabilities.
- Morocco is a major nearshore sourcing destination for Europe, offering lower CMT costs, flexible MOQ production, fast-fashion efficiency, and shorter EU lead times.
- The selected oversized heavyweight hoodie is highly commercial due to rising streetwear demand, Gen Z fashion trends, athleisure growth, and premium casualwear positioning.
- Turkey is more suitable for premium and sustainable apparel sourcing, while Morocco is more suitable for fast-fashion and replenishment manufacturing.
- Both countries remain strategically important sourcing hubs for European apparel brands.

